

Gold Pipeline v1.4.13 — Live Fix Verification Report

Re-run of the weak / medium / strong stress-test essays on the fixed pipeline (engine_version 1.4.13-evaluator-before-scorer-rubric-bridge). All three sessions completed with qa_status: passed and zero stage failures.

VERDICT: NOT YET READY FOR COMMERCIAL RELEASE

7 of 9 previously-documented bugs are confirmed fixed against live, real data with zero regressions. But this verification pass surfaced a new, concrete scoring-accuracy defect: the medium and strong essays — two essays deliberately written to be clearly different quality — now receive **identical** final scores (5.5 overall, same criteria bands across the board). The root cause is traced below. Given the product's core value proposition is accurate differentiated scoring, this blocks release until fixed.

1. Confirmed fixed — verified against live run data

Problem	Status	Live evidence
1 — Practice engine criterion mismatch	FIXED	Strong essay: 11/11 exercises delivered (was 0/11). Weak/medium/strong exercise family lists are all distinct.
2 — Scorer TR/CC flat at 0.55	MOSTLY FIXED	task_response now monotonic: 0.528 → 0.631 → 0.643 (weak→medium→strong). grammar also differentiates (0.530 → 0.576 → 0.581). coherence_cohesion differentiates but isn't perfectly monotonic (0.837 / 0.857 / 0.819) — a known, previously-flagged limitation.
3 — Damage signals not wired	FIXED	local_language_damage_index: 1.0 / 0.18 / 0.09 (weak/medium/strong). serious_error_sentence_ratio: 0.67 / 0.09 / 0.0 — correctly reflects Problem 4's fix removing strong's one false-positive row.
4 — det_vip CLAUSE_STRUCTURE false positive	FIXED	The "background, is" false positive on the strong essay's modal-passive sentence is gone from the detector output entirely.
5 — criteria_math_valid always false	FIXED	criteria_math_valid: true on all three essays. The governor-repair safety net didn't even need to fire this run (criteria_preserved: true) — the upstream numbers were already consistent.
6 — LRET v1110 modal-passive false positives	FIXED	Zero v1110 grammar-flag clarify_units across all three essays (previously medium: 1, strong: 2, both on correct passives like "should be mandated", "cannot be manufactured").
7 — LRET FIX confidence bar (civic equity case)	INCONCLUSIVE	fix_units on the strong essay is now 0 (was 1). But "civic equity" itself didn't reach the FIX pipeline this run at all — it was blocked upstream and classified KEEP, so the demotion logic never got exercised. Code fix is in place and logic-verified, but this specific scenario didn't reproduce live (LLM-pass non-determinism).
8 — Writing Coach micro_lesson ships empty	FIXED	"do" is populated on all three essays. "avoid" is populated on strong (the documented case) and empty on weak/medium — expected, since only 13/44 moves in the bank have extractable avoid content; confirmed as honest behavior, not a regression.
9c — Revision engine contradictory guidance	FIXED	Introduction paragraph function_alerts is now empty on both weak and strong (previously contradicted the "do not add examples" hint). The suppressed text is logged in evidence_sources for audit.

2. New finding: lexical_resource never differentiates (release blocker)

raw_criterion_quality.lexical_resource is **exactly 0.5645** on all three essays — weak, medium, and strong. This is the same class of bug as Problem 2 (a criterion stuck on a flat default because its real value never gets written into the scorer's highest-priority metric source), just never addressed by the evaluator_rubric_bridge fix, which only built task_response and coherence_cohesion sub-metrics.

Traced directly: scorer_payload.premium_metric_profile_mapped_metrics has no lexical_resource key at all on any of the three essays, and detector_metric_profile.lexical_resource is an empty {} on all three. With no real signal at any priority level, the scorer is falling back to a hardcoded default for this one criterion, for every essay, regardless of actual lexical quality.

Downstream impact: this is very likely why the medium and strong essays collapsed to an identical final score. Their raw_criterion_quality for task_response (0.631 vs 0.643) and coherence_cohesion (0.857 vs 0.819) are close but not equal — under normal rounding that alone could still separate them. But with lexical_resource pinned flat at 0.5645 → grammar 5 vs 5 and lexical_resource 5 vs 5 for both, the two essays landed on identical criteria_bands (task_response 6, coherence_cohesion 6, lexical_resource 5, grammar 5) and an identical overall_band of 5.5.

Recommended fix: extend evaluator_rubric_bridge_v1.py with an LR1-style rubric built from the Evaluator's Lexical Control / Advanced Lexical Competence skill_observation_profile entries, the same way task_response and coherence_cohesion were built for Problem 2 — then re-verify that medium and strong separate on lexical_resource and, ideally, on overall_band.

3. Score outcome comparison

Essay	task_response	coherence_cohesion	lexical_resource	grammar	Final overall_band
Weak	0.528	0.837	0.5645	0.530	4.5
Medium	0.631	0.857	0.5645	0.576	5.5
Strong	0.643	0.819	0.5645	0.581	5.5 ← same as medium

lexical_resource's column is identical top to bottom (0.5645/0.5645/0.5645) — the tell that this criterion carries zero real per-essay signal, unlike the other three columns which all move in a sensible direction.

4. Regression check

All three sessions: qa_status passed, zero missing/invalid required artifacts, zero boundary or quality-gate issues, zero stage failures, zero stderr output in any log. No evidence of anything the fixes broke.

5. Items intentionally left open (not new, not blockers on their own)

Problem 7's deeper concern — that a FIX suggestion can be surface-correct but drop a deliberate argumentative thread — remains an open design question, not a code bug. Problem 9a/9b's "no sentence ever reaches green" behavior is confirmed working as designed: green requires simultaneous agreement between the Evaluator's independent judgment and the Detector's chargeable-row count, and that's a strict, intentional bar, not a defect. Writing Coach's specific skill-selection choice on the strong essay (formality_control) traces to a legitimate multi-signal mastery calculation, not a bug.

6. Path to release

Build the missing lexical_resource rubric in evaluator_rubric_bridge_v1.py (mirrors the already-proven Problem 2 pattern), re-run the three stress essays, and confirm medium and strong separate on both lexical_resource and overall_band. That is the one remaining concrete blocker identified in this pass.